

ETHAN (TAEHOON) LEE

US Permanent Resident · ethantlee21@gmail.com · thdxg.dev · linkedin.com/in/ethantlee · github.com/thdxg

EDUCATION

Carnegie Mellon University

Master's in Software Engineering - Scalable Systems

Pittsburgh, PA

Expected Dec 2027

Emory University

BS in Computer Science

Atlanta, GA

May 2026

- **Honors:** 3.99/4.0 GPA, Magna Cum Laude, 2x James T Laney Scholarship, 5x Dean's List, Phi Beta Kappa Honor Society

SKILLS

- **Languages:** Go, TypeScript, JavaScript, Java, C, Rust, Python, SQL, HTML, CSS, x86 Assembly
- **Technologies:** Kubernetes, Linux, Docker, AWS, GCP, Cloudflare, PostgreSQL, Redis, OpenTelemetry, Temporal

WORK EXPERIENCE

Software Engineer

Huddle Surety

Atlanta, GA

Sep 2025 – Present

- Innovating the surety industry with an AI-powered bond automation platform reducing the bond processing time by 90%
- Built multi-stage RAG pipelines with cross-document context retrieval generating bond documents with 95% accuracy
- Developed the core Go REST API spanning 100+ endpoints with Redis caching, relationship-based auth, and audit logs
- Shipped a role-based frontend delivering distinct surety agent and contractor UX using Next.js, TypeScript, and tRPC
- Architected an OpenTelemetry observability pipeline with end-to-end context propagation and tracing across services

Software Engineering Intern

UKG

Atlanta, GA

May 2025 – Aug 2025

- Developed a production REST API in Go implementing the worker pool concurrency pattern to aggregate and monitor network health of 200+ MongoDB, MySQL, and RabbitMQ servers across 4 geodistributed datacenters real-time
- Implemented a TLS certificate validator using Kubernetes CronJobs, accelerating the biannual renewal process by 70%
- Provisioned Prometheus Blackbox Exporters across isolated OpenStack tenants to achieve distributed observability

Software Engineering Intern

eStreamly

Atlanta, GA

Jun 2024 – Aug 2024

- Optimized global state management and client-side caching using TypeScript and Angular, reducing page latency by 40%
- Created multi-stage container build and release pipeline on GCP Cloud Build, improving deployment speed by 35%

PROJECTS

Macterm: MacOS Terminal Emulator with Persistent Multiplexing

github.com/thdxg/macterm

- Built a native macOS terminal with SwiftUI and libghostty, gaining 5K+ downloads and 15 contributors globally
- Implemented session persistence, declarative layout configuration, fuzzy command palette, and configurable hotkeys

Kubernetes Homelab

github.com/thdxg/homelab

- Designed a two-node Kubernetes cluster on Raspberry Pi, fully automating CI/CD via GitOps using FluxCD and sops
- Published personal projects and dashboards via Gateway API ingress with automatic TLS certificates and dynamic DNS
- Contributed to Kubernetes documentation on pod scheduling and eviction by synchronizing outdated Korean docs
- Contributed to Cilium CNI by tracing a runaway loop with livelock causing high CPU usage and silent traffic drops

Spotify Warped: HCI Research & Publication

dl.acm.org/doi/10.1145/3772318.3790875

- Published to ACM CHI 2026 on how Spotify's algorithmic feedback loops shape listening behavior and data engagement
- Built high-throughput data processing pipelines in Python to ingest 100 GB of Spotify streaming data from 60 users

Emory Hacks: Emory University's Hackathon Organization

emoryhacks.com

- Co-founded and developed annual hackathon website and registration system using Next.js, Spring Boot, and Firebase